Program 2 : Develop a multistage docker file for container orchestration

Step 1 : mkdir program2
cd program2
nano Dockerfile

```
FROM node:20-alpine AS builder
WORKDIR /app
COPY package.json package-lock.json ./
RUN npm install
COPY . .
RUN npm run build


FROM node:20-alpine
WORKDIR /app
COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules
EXPOSE 3000
CMD ["node","dist/index.js"]
```

Step 2 : npm init -y
npm install

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/program2$ npm init -y
Wrote to /home/1rv24mc040-haripriya/Devops_lab/program2/package.json:

{
  "name": "program2",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}


1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/program2$ npm install

up to date, audited 1 package in 115ms

found 0 vulnerabilities
```

.

Step 3 : mkdir src
cd src
nano index.js

```
const express=require("express")
const app=express();
const PORT=3000;
app.get('/',(req,res) => {
        res.send("Hello from Docker");
});
app.listen(PORT,() =>{
        console.log(`Server listening on port ${PORT}`);
});
```

Step 4 : docker build -t program2 .

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/program2$ docker build -t program2
 .
[+] Building 5.8s (15/15) FINISHED                                docker:default
 => [internal] load build definition from Dockerfile                       0.0s
 => => transferring dockerfile: 419B                                       0.0s
 => [internal] load metadata for docker.io/library/node:20-alpine          1.6s
 => [internal] load .dockerignore                                          0.0s
 => => transferring context: 2B                                            0.0s
 => [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b97  0.0s
 => [internal] load build context                                          0.0s
 => => transferring context: 483B                                          0.0s
 => CACHED [builder 2/6] WORKDIR /app                                      0.0s
 => [builder 3/6] COPY package.json package-lock.json ./                   0.0s
 => [builder 4/6] RUN npm install                                          3.3s
 => [builder 5/6] COPY . .                                                 0.0s
 => [builder 6/6] RUN npm run build                                        0.4s
 => [stage-1 3/6] COPY --from=builder /app/package.json ./                 0.0s
 => [stage-1 4/6] COPY --from=builder /app/package-lock.json ./            0.0s
 => [stage-1 5/6] COPY --from=builder /app/dist ./dist                     0.0s
 => [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules     0.1s
 => exporting to image                                                     0.1s
 => => exporting layers                                                    0.1s
 => => writing image sha256:15f269c0d733507d65f4d4a7b734970d7f51abc45db29  0.0s
 => => naming to docker.io/library/program2                                0.0s

View build details: docker-desktop://dashboard/build/default/default/xkd2o2hlwyoa0p
ah4wfa2ds45
```

.

Step 5 : docker run -p 3000:3000 program2

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/program2$ docker run -p 3000:3000
program2
Server listening on port 3000
```

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Step 6 : Check the output in the browser
http://localhost:3000